Pratik Kailas Patil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.60
ACS college dharangaon	Class XII (Maharashtra state board)	2019	79.85%
S D Kude high school dharangaon	Class X (Maharashtra state board)	2017	93.60%

SKILLS

Simulation : STAR CCM +,ANSYS Workbench,ICEM CFD,ANSYS Fluent .

Numerical Methods : Finite Volume Method **(FVM)**,Finite Difference Method**(FDM),**Finite Element Method (FEM).

Programming: MATLAB.

CAD :CATIA V5.

PROJECTS

1. M.Tech Projects

Novel Data-Driven Methods for Optimization of Arrangement of Multiple Cylinders in Fluid Flow to Enhance Hydrodynamic Forces

M.Tech Final Year Project | Ongoing

- Developing a data-driven CFD framework to investigate and optimize the arrangement of multiple circular cylinders for enhanced hydrodynamic forces.
- Performing 2D unsteady incompressible flow simulations using **Siemens STAR-CCM+** to analyze wake dynamics, vortex shedding, lift and drag characteristics.
- Conducting mesh-independence and time-step studies to ensure numerical accuracy and reliable prediction of transient flow behavior.
- Processing CFD flow-field and force-coefficient data using **MATLAB** for data-driven analysis and optimization.

M.Tech Course Projects

1)Numerical Simulation of 2D Unsteady Diffusion Using FVM

- Developed a Finite Volume Method solver and implemented **Explicit Euler, Implicit Euler and Crank-Nicolson** time-integration schemes.
- Compared the schemes based on stability, accuracy and solution behavior.

2)CFD Analysis of NACA 0012 Airfoil

- Generated structured mesh using **ANSYS ICEM CFD** and performed CFD simulations in **ANSYS Fluent**.
- Investigated pressure and velocity distributions at **3°, 9° and 16° angles of attack**, including flow separation and wake formation.

3)Lid-Driven Cavity Flow Using SIMPLEC

- Developed a **MATLAB-based finite-volume solver** using the SIMPLEC algorithm for incompressible cavity flow.
- Studied flow behavior at **Re = 100, 400 and 1000** using velocity fields, streamlines and centreline profiles.

4)Supersonic Nozzle Design and CFD Validation

- Designed a supersonic nozzle contour using the **Method of Characteristics (MOC)** and Prandtl-Meyer relations.
- Performed CFD validation and compared the numerical flow characteristics with analytical predictions.

5)Shock-Tube Flow: Analytical and CFD Investigation

- Investigated shock-tube flow using analytical relations and CFD simulations.
- Compared **pressure, velocity, temperature, Mach number and shock propagation** between analytical and CFD results.

6)2D CFD Modelling of H-Darrieus Vertical Axis Wind Turbines Using a Transition Turbulence Model

- Developed transient CFD simulations using the **Sliding Mesh Model** and Transition SST turbulence model.
- Calibrated turbulence-model correlation parameters using a C-based User-Defined Function (UDF) to improve lift and drag predictions.
- Conducted mesh-independence and time-step-independence studies to ensure numerical reliability.
- Validated aerodynamic coefficients against experimental data and compared NACA 0015 and NACA 4518 rotor performance.
- Analysed power coefficient , tip-speed ratio , dynamic stall, and wake-flow characteristics.
- Achieved **6-8% deviation in predicted power coefficient** from experimental data using the calibrated Transition SST model.

2. B.E. Projects

B.E. Final-Year Project

Machining Cost Estimation Using AI

B.E. Final Year Project | Oct 2022 - May 2023

- Developed an automated **machining cost-estimation and quotation system** for processing client requests and CAD file attachments.
- Implemented AI-based machining feature detection using **SSDNet** to identify manufacturing features from CAD models.
- Automated generation of **PDF quotations** and structured logging of quotation data to improve the efficiency of the costing process.

B.E. Course / Academic Projects

1)Design of a Multispeed Multistage Gearbox

- Designed a **four-speed, two-stage gearbox** for an input speed of 3000 rpm and output speed range of 1100-3400 rpm.
- Determined gear ratios and performed gear design calculations for the specified power-transmission requirements.

2)Design of a Flexible Bush Coupling

- Designed a **55 kW flexible bush coupling**, including shaft, flange, hub, key, pins and bushes.
- Performed component sizing and strength checks based on torsional strength and permissible-stress criteria.

EXTRACURRICULAR

SUPRA Student Team - Chassis Team

- Worked as a member of the **Chassis Team** during B.E., contributing to chassis-related design and development activities for the student formula racing team.

- Participated in the **Formula Bharat Virtual Event**, gaining exposure to vehicle design, engineering practices, and technical team collaboration.

- Developed hands-on familiarity with **CATIA** for 3D modelling and chassis-related design work.

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Heat Transfer & Fluid Mechanics Laboratories

- Assisted in conducting and supervising Heat Transfer Laboratory sessions for 3rd-year B.Tech. students and Fluid Mechanics Laboratory sessions for 2nd-year B.Tech. students.

- Guided students in experimental procedures, equipment operation, data acquisition, calculations, and interpretation of results.

- Supported students in understanding fundamental concepts and resolving experimental and technical queries.

RELEVANT COURSES

MTECH COURSES : Advanced Computational Fluid Dynamics (CFD), Computational Fluid Dynamics, Turbulence, Advanced Heat Transfer, Incompressible Fluid Flow, Compressible Flow, Thermal Fluid Laboratory, CFD Laboratory, Mathematics Tools for Engineers, Machine Learning.

BE COURSES: Basic Thermodynamics, Power Plant ,HVAC ,Fluid mechanics,Heat transfer, Turbo Machinery, Strength Of Materials ,Machine Design,Manufacturing Process.